Introduction

Parkinson's Motor Impairment

- ★ **Parkinson's disease (PD)** impairs over **10 million people**[1], costing the U.S. **$52 billion annually**[4]
  - **Incurable & rising** amongst aging global cases[1]
  - Disrupts muscle activity → **morbidity & mortality**[2]
  - Caused by loss of **dopamine**-producing neurons[3]
- ★ **Bradykinesia & muscular degeneration**
  - Restricts daily activities → **pain & dependency**[2]
  - **Major gap** in symptom monitoring[5]

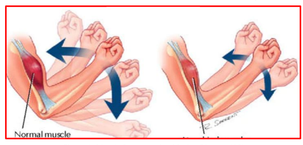

Figure 1: Muscle Degeneration Motion Diagram(16)

Muscular Degeneration

- ★ **Occurs across all stages of PD**[6]
  - **Progressively evolves**[6] → solutions must adapt
- ★ **Sarcopenia**[3] & **fatigue**[2] limit patient motion
  - Predominantly **lower-limbs**[7] **& elbows**[10]
- ★ Linked to **mitochondrial dysfunction**[8] (low muscle power) & **lack of physical activity**[2]
- ★ **Potential** of assistive devices in restoration[9]

$$F_{deg} = F_{max} \cdot \alpha_1 \cdot e^{-(\alpha 2+\beta t)} = \text{m_loss} \cdot \text{m_fati}$$

Equation from Myosuite (Caggiano et al.) (18)

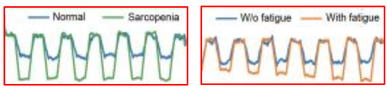

Figure 2-3: Myosuite's Sarcopenia + Fatigue on Muscle Activations (18)

Bradykinesia

- ★ Affects **98%** of PD patients[12]
- ★ Significant **slowness** & delay (**akinesia**) of movement, often asymmetrical[14] → **fall risk**
  - **10-40%** slower[13]
  - Mental challenges: **frustration**[17]

$$N_{brady} = x(i\text{-}\Delta i) \cdot \alpha = \text{delay \& slowness}$$

Bradykinesia Equation Created with Research from (17)

- ★ Caused by **basal ganglia dysfunction**[14]
- ★ Sensors (**IMU**) can assess severity[15]

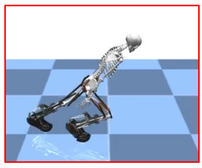

Figure 4: Simulated PD patient - Created by Researcher

Limitations

[1] - Adaptive nature of RL & randomized simulator
[2] - Low-cost, accessible product
[3] - High-fidelity, validated simulator
[4] - Intelligent **EMG/IMU** analysis for movement & impairment prediction
[5] - Wide-ranging set of exoskeletons
[6] - Cloud Computing Pipeline

Implemented in theory, not in practice → lacks real-time & patient-adaptive functionality.

Patient Specific	[1][5]	Expensive	[2][6]
Data Overfitting	[1][3]	Manual Control	[4][6]
Invasive + Side Effects	[3][4]	Simplified mechanics	[3][4]

Research Objective

Research Question
Can machine learning systems achieve real-time, adaptive mitigation and tracking of Parkinson's motor impairments across diverse patients?

Criteria

- ★ **Real-time functionality**
  - **Seamless**, natural, effective interaction
- ★ **Adaptive** → **High restoration %** across all patient PD levels & movements
- ★ **Safe, non-invasive**
- ★ User-friendly → **intuitive** device

Constraints

- ★ **Real patient data unavailable** → sim-to-real transfer
- ★ **Low-cost** exoskeleton → hardware & model limitations
- ★ Limited **computational resources**
  - Lenovo PC hardware (16GB memory)

Engineering Goal

Engineering Goal: Use Reinforcement Learning (RL) to create an adaptive, low-cost solution for motor impairments from Parkinson's Disease

Primary Goal
Provide real-time assistance (**<250ms**) with **over 90%** movement restoration

Secondary Goal
Monitor impairment progression with an assistance correlation (r) **above 0.95**

*ReGainX assists across all stages and movement, providing the **first exoskeleton system** tailored to **all** PD patients*

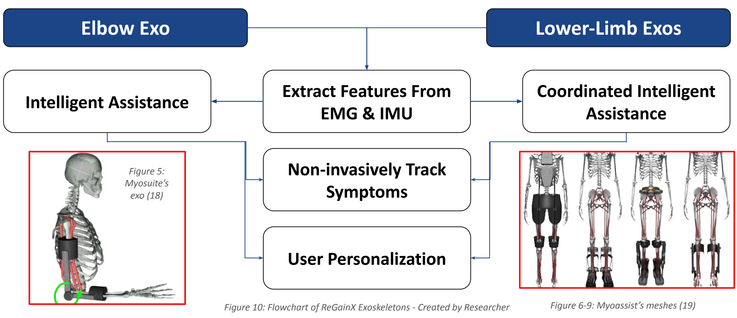

Figure 5: Myosuite's exo (18)

Figure 10: Flowchart of ReGainX Exoskeletons - Created by Researcher

Figure 6-9: Myoassist's meshes (19)

Engineering Methodology

Design Approach

- ★ <u>**Reinforcement Learning**</u>
  - Learning through **trial & error**
    - **Markov decision process**
    - Actions → Outcomes → Rewards/Penalties
  - **Superior to traditional AI at tasks humans don't know the answer to**
  - Excels at complex control systems[23]

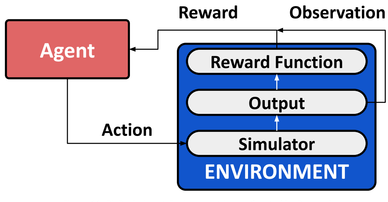

Figure 11: RL Basic Agent-Environment Loop - Created by Researcher

Alternative Solutions Explored
Smart-cueing footwear, AI-controlled dosing, VR rehabilitation system, manual exoskeletons, EMG feedback band.

```
action,_ = exo.predict(obs)
obs,reward,done, _ = env.step(action)
```

Figure 12: Markov Decision Process – Created by Researcher